Neutrino Oscillations in KM3NeT

Course Project Report

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Course: Theoretical Astroparticle Physics (PHYS 708)

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1 Introduction

This report is a literature review that I conducted as part of a course project in astroparticle physics. As a student with a background and interest in statistics, my aim was to learn something new about astroparticle physics while also trying to connect it to statistical ideas. At first, my plan was to focus mainly on the statistical techniques used in the field and to talk about those in some detail. However, I quickly realized that there is a lot of necessary physical and theoretical background that needs to be understood before one can properly discuss statistical methods and data analysis in this context. In the process of working through this background, I found the subject of neutrino oscillations to be particularly fascinating. Because of this, I decided to concentrate my literature review on the theoretical and conceptual background required to understand neutrino oscillations, rather than on detailed statistical techniques themselves. The result is this report, which is mainly focused on providing that background, and I hope that it can be helpful for future students who are entering the field and who might be trying, like I was, to connect their interests in mathematics and statistics with topics in astroparticle physics.

2 History of Neutrinos

The story of the neutrino begins with Pauli in 1930. Faced with an apparent violation of energy–momentum conservation in beta decay, he proposed a neutral, nearly undetectable particle to fix the puzzle[1]. A few years later, in 1933, Fermi developed the theory further and coined the name "neutrino," the "little neutral one." [2] Only in 1956 did Reines and Cowan finally detect antineutrinos through inverse beta decay, turning a conjecture into a measured particle.[3] Since then we've learned a great deal about neutrinos—their properties and how they interact—but there is still plenty to uncover about their place in the Standard Model and their role in broader physical phenomena.

2.1 Neutrino Properties

Particle physics today is organized by the Standard Model, a remarkably successful framework for describing known particles and the forces acting on them. It is our closest attempt at a unified description of matter and interactions, yet it has gaps. Different communities probe different sectors—bosons and leptons among them—to push on those gaps and expose the deeper physics underneath. Neutrinos are central in that effort. Neutrino oscillations, in particular, sit uneasily with the minimal Standard Model and remain a major driver of current research aimed at clarifying how neutrinos actually fit into the theory.

Within the Standard Model, neutrinos are leptons with no electric charge and no color charge. As a result, they do not participate in electromagnetic or strong interactions. Their dominant interactions are weak, mediated by the neutral Z^0 and the charged W^+/W^- bosons.

In the Standard Model, neutrinos were treated as massless. Evidence from neutrino oscillations—flavor change during propagation—indicates otherwise, implying non-zero masses. This tension between the original model and observation compels us to probe the fundamental nature of neutrinos more closely.

Like their charged lepton partners (electrons, muons, taus), neutrinos come in three flavors. These flavors are not the same as mass states, and that mismatch is what allows a neutrino produced as one flavor to be detected later as another. The first clues came from the solar neutrino problem in the late 1960s, when fewer electron neutrinos were measured than expected. Super-Kamiokande later observed a zenith-angle–dependent disappearance of atmospheric muon neutrinos, strengthening the oscillation picture. By 2002, measurements demonstrated that the Sun's missing electron-neutrino flux reappears as other flavors through weak interactions. Together, these results established that neutrino oscillations occur—and that neutrinos have mass.

3 Neutrino Interactions & Types

3.1 Neutrino Interactions with Matter

Neutrinos interact with matter exclusively through the weak interaction, which gives them extremely small interaction probabilities but produces distinctive observable signatures in large-volume detectors. There are two fundamental weak-interaction channels: charged-current (CC) and neutral-current (NC) interactions.

In a charged-current interaction, a neutrino exchanges a W boson with a target electron or nucleon, producing a charged lepton of the same flavor:

$$\nu_\ell + N \to \ell^- + X, \qquad \ell = e, \mu, \tau.$$

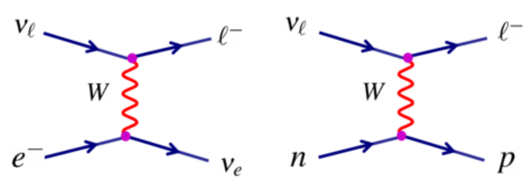

Figure 1: Feynmann Diagram of a Current Charged Interaction [4]

Charged-current reactions are flavor-specific and typically deposit large amounts of energy in the detector, yielding clear topologies such as electromagnetic showers or long muon tracks.

Neutral-current interactions proceed through the exchange of a Z boson:

$$\nu_\ell + N \to \nu_\ell + X,$$

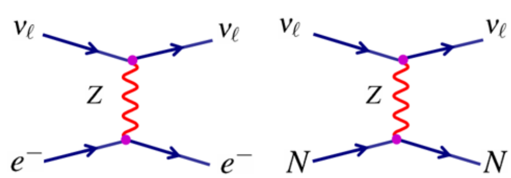

Figure 2: Feynmann Diagram of a Neutral Charged Interaction [4]

and are flavor-blind. Because the outgoing neutrino carries away an unknown fraction of the energy, NC events generally produce dimmer, more compact hadronic cascades.

Charged-current interactions therefore play a central role in neutrino oscillation studies and astrophysical analyses, as they provide both flavor information and higher visible energy. Neutral-current interactions, although less informative individually, are crucial for measuring the total neutrino flux independent of flavor composition.

3.2 Energy Thresholds for Neutrino Interactions

Charged–current (CC) neutrino interactions require sufficient energy to produce a charged lepton of mass m_ℓ. This leads to well-defined kinematic thresholds that depend on both the target particle and the neutrino flavor. These thresholds play a central role in determining which neutrino flavors can be detected at different energies in experiments such as Super-Kamiokande, KM3NeT, and IceCube. Neutral–current (NC) interactions, by contrast, do not produce charged leptons and therefore do not have a lepton-mass threshold. [5]

CC Threshold on Electrons

Consider the reaction $\nu_\ell + e^- \to \ell^- + \nu_e$ in the lab frame, where the electron is initially at rest. The Mandelstam invariant is

$$s = (p_\nu + p_e)^2 = m_e^2 + 2m_e E_\nu,$$

since $p_\nu^2 \simeq 0$ for a nearly massless neutrino. At threshold, the invariant mass of the system must just equal the mass of the charged lepton produced:

$$s_{\text{thr}} = m_\ell^2.$$

Equating and solving for the neutrino energy,

$$m_e^2 + 2m_e E_\nu = m_\ell^2 \quad \Rightarrow \quad E_\nu = \frac{m_\ell^2 - m_e^2}{2m_e} = \left[\left(\frac{m_\ell}{m_e}\right)^2 - 1\right]\frac{m_e}{2}.$$

This expression shows explicitly why producing heavier leptons requires dramatically higher neutrino energies.

CC Threshold on Nucleons

For quasi-elastic scattering on nucleons, such as $\nu_\ell + n \to \ell^- + p$, the neutron is again at rest in the lab frame. The invariant is

$$s = m_n^2 + 2m_n E_\nu.$$

At threshold, the outgoing proton and charged lepton appear with no relative momentum in the center-of-mass frame, so

$$s_{\text{thr}} = (m_p + m_\ell)^2.$$

Thus,

$$m_n^2 + 2m_n E_\nu = (m_p + m_\ell)^2 \quad \Rightarrow \quad E_\nu = \frac{(m_p^2 - m_n^2) + m_\ell^2 + 2m_p m_\ell}{2m_n}.$$

This expression gives thresholds of order MeV for electron neutrinos (inverse beta decay), hundreds of MeV for muon neutrinos, and a few GeV for tau neutrinos.

Numerical Thresholds and Experimental Implications

The numerical thresholds derived from these formulas are summarized in Table 1. Electron neutrinos can be detected at arbitrarily low energies on electrons and at MeV energies on nucleons. Muon-neutrino CC interactions require energies above $\sim 10^2$ MeV (nucleons) or several GeV (electrons). Tau-neutrino CC channels demand multi-GeV energies. These thresholds explain why tau appearance requires high-energy atmospheric or astrophysical neutrinos and why detectors operating at MeV scales (e.g. solar neutrino detectors) observe only electron-flavor CC events.

NC interactions have no charged-lepton threshold because the outgoing neutrino remains invisible; the only visible energy comes from the recoiling hadronic system. As a result, NC events occur at all energies above the nuclear breakup scale but cannot be used for flavor identification. [4]

Table 1: Charged–current energy thresholds for neutrino interactions on electrons and nucleons.

Neutrino Flavor	E_{th} (e^- target)	E_{th} (nucleon target)
ν_e	0	0
ν_μ	11 GeV	110 MeV
ν_τ	3090 GeV	3.5 GeV

These thresholds fundamentally shape which neutrinos are observable in each experiment and define the scientific reach of modern neutrino telescopes.

From the perspective of weak-interaction detection, neutrino oscillations appear as a flavor-dependent suppression in charged-current event rates. Solar detectors primarily observe ν_e through charged-current interactions. As solar electron neutrinos propagate from the Sun to Earth, they oscillate into ν_μ and ν_τ, which are not efficiently detected at low energies due to the energy thresholds for muon and tau production. The result is an apparent deficit of solar ν_e in CC-sensitive detectors.

Atmospheric neutrino detectors observe ν_μ disappearance for similar reasons. Upward-going atmospheric muon neutrinos traverse long baselines through the Earth and oscillate primarily into ν_τ. Because tau production requires energies above several GeV, many oscillated ν_τ do not yield visible tau leptons. The charged-current ν_μ rate is therefore strongly suppressed for certain energies and path lengths.

Neutral-current event rates remain unchanged across these oscillation phenomena, since NC interactions are flavor-blind. This is why NC measurements, such as those performed by SNO, confirm that the total neutrino flux is conserved even when the flavor composition changes.

3.3 Neutrino Cross Sections and Interaction Rates

Neutrino interaction cross sections are extremely small because the weak force has a short range and couples only weakly to matter. As the neutrino energy increases, the cross section grows approximately linearly in the high-energy regime. For charged–current (CC) processes the dominant contribution scales as

$$\sigma_{\mathrm{CC}} \propto G_F^2\, m_T\, E_\nu,$$

where m_T is the target mass (m_e for electrons, m_p for nucleons) and G_F is the Fermi constant. Scattering on electrons therefore yields much smaller energy transfers than scattering on nucleons, which produce hadronic debris and significantly larger visible energy. CC interactions involve W-boson exchange and typically have larger cross sections than neutral–current (NC) interactions, which proceed via Z exchange.

The characteristic orders of magnitude of neutrino cross sections across relevant energy scales are consistent with the behavior shown in Fig. 3:

- **Solar and reactor neutrinos (MeV scale):** $\sigma \sim 10^{-45} - 10^{-43}$ cm^2.
- **Atmospheric and accelerator neutrinos (GeV scale):** $\sigma \sim 10^{-39} - 10^{-38}$ cm^2.
- **Astrophysical neutrinos (TeV–PeV scale):** $\sigma \sim 10^{-36} - 10^{-35}$ cm^2.

The rapid rise of $\sigma(E_\nu)$ with energy is central to the operation of high-energy neutrino telescopes such as KM3NeT/ARCA

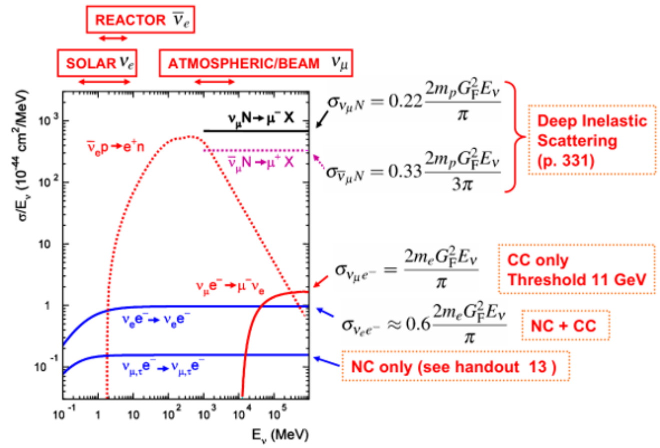

Figure 3: Crossection Formula for Varying Neutrino Processes [4]

and IceCube. Although neutrinos remain weakly interacting even at PeV energies, the increased cross section enables kilometer-scale detectors to observe astrophysical neutrino fluxes with statistically significant event rates.

3.4 Neutrino Types: Solar, Atmospheric, Cosmological, and Cosmogenic

Neutrinos originate from a variety of natural sources spanning an enormous range of energies.
Solar neutrinos are produced in nuclear fusion processes within the Sun, primarily through the *pp* chain, ^{8}B decay, and the CNO cycle. These neutrinos have energies from 0.1 to 20 MeV and produce an enormous flux at Earth,

$$\Phi_{\nu_\odot} \approx 2 \times 10^{10}\ \nu\ \text{cm}^{-2}\ \text{s}^{-1}.$$

They are almost entirely electron neutrinos due to the charged-current weak processes of solar burning.

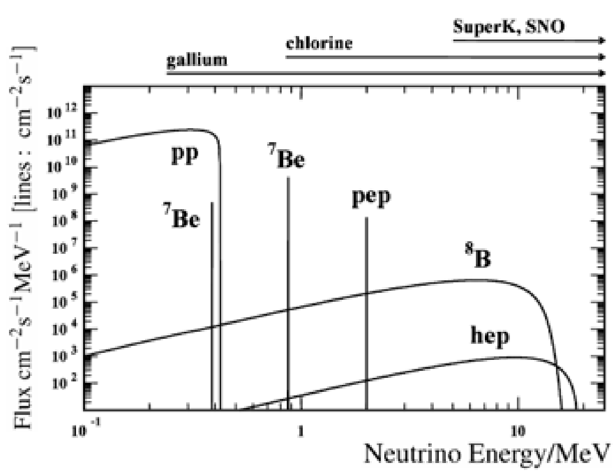

Figure 4: Solar Neutrino Flux of different processes. [6]

Atmospheric neutrinos arise when cosmic rays strike the upper atmosphere, producing charged pions that decay via

$$\pi^\pm \to \mu^\pm + \nu_\mu, \qquad \mu^\pm \to e^\pm + \nu_e + \nu_\mu.$$

These neutrinos span energies from hundreds of MeV to beyond 100 TeV and are initially dominated by ν_μ. Their propagation through the Earth makes them ideal for studying oscillations and matter effects.

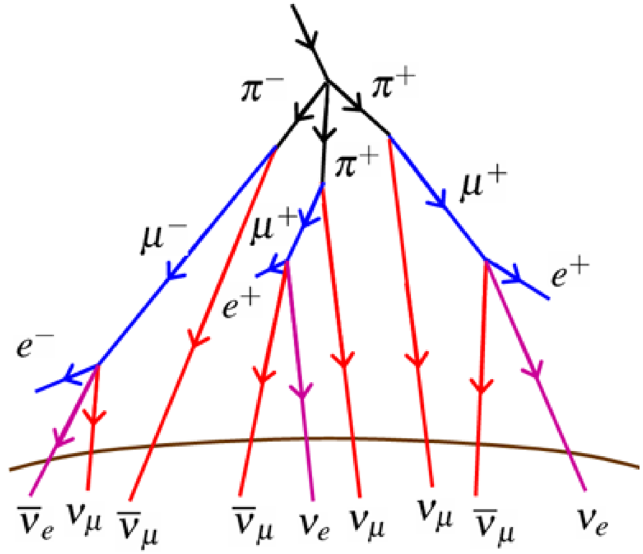

Figure 5: Cosmic Ray cascade forming atmospheric neutrinos [4]

Cosmogenic neutrinos (also known as GZK neutrinos) are produced when ultrahigh-energy cosmic rays interact with cosmic microwave background photons. Their energies can reach 10^{17}–10^{19} eV, but their flux is extremely low, motivating the use of multi-kilometer-scale detectors.

Cosmological neutrinos, the relics of the Big Bang, have energies near 10^{-4} eV. Although currently undetectable, they play a profound role in cosmology and structure formation.

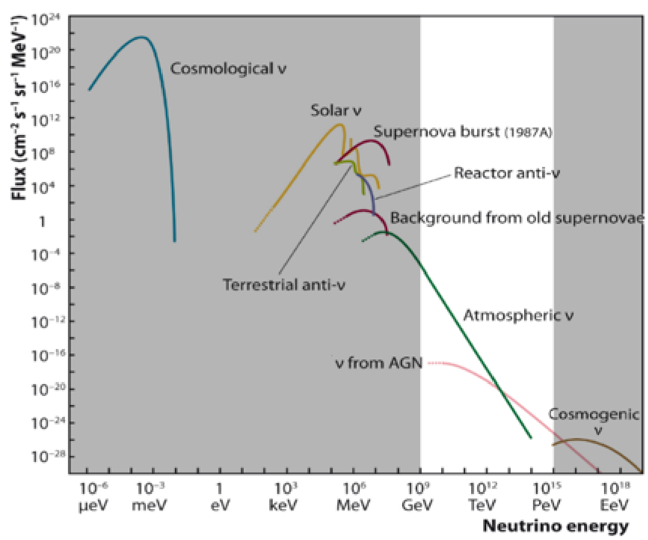

Figure 6: Neutrino Energy versus flux for different neutrino types. White band range represents the energy rang interest of KM3NeT. [7]

3.5 Neutrino Event Topologies in Detectors

When neutrinos interact in large water or ice Cherenkov detectors, they produce characteristic patterns of Cherenkov light—referred to as *event topologies*—that depend on the neutrino's flavor, energy, and interaction channel. These topologies allow experiments such as KM3NeT/ORCA, KM3NeT/ARCA, IceCube, and Super-Kamiokande to reconstruct the direction, energy, and sometimes the flavor of the incoming neutrino.

In charged–current (CC) interactions, the neutrino is converted into its corresponding charged lepton, and the lepton's propagation largely determines the observable signature. In neutral–current (NC) interactions, the outgoing neutrino remains undetected, and only the hadronic recoil is visible. The clarity of these signatures is further influenced by detector geometry—particularly the spacing of Digital Optical Modules (DOMs)—and by the optical properties of the medium. Deep-sea water has long scattering lengths and low absorption, yielding especially sharp Cherenkov patterns and excellent angular resolution for track-like events. [7]

Event Topology by Neutrino Type

- **Neutral–Current (NC) Interactions**
  NC interactions of any flavor produce only a hadronic shower, since the outgoing neutrino escapes undetected. The resulting cascade is generally compact and carries less visible energy than the true neutrino energy. Although NC events do not contain flavor information, they contribute to measurements of the total neutrino flux and are essential for oscillation studies that compare CC and NC rates.

- **Charged–Current Electron Neutrinos (ν_e CC)**
  A ν_e CC interaction produces an energetic electron or positron that rapidly initiates an electromagnetic (EM) shower. Electrons travel only ~ 36 cm in water before losing most of their energy, and the developing EM cascade extends over only a few meters. These events appear as bright, approximately spherical Cherenkov light patterns and have excellent energy resolution. Because EM showers emit about 20% more light than hadronic showers of the same energy, ν_e CC events are distinguishable from NC cascades.

- **Charged–Current Muon Neutrinos (ν_μ CC)**
  A ν_μ CC interaction produces a muon, a long-lived charged lepton whose range may extend hundreds of meters at GeV–TeV energies. As the muon traverses the detector, it emits a straight, extended Cherenkov track. These track-like events yield the best angular resolution of any topology, making ν_μ CC interactions the primary channel for neutrino astronomy in KM3NeT/ARCA and IceCube.

- **Charged–Current Tau Neutrinos (ν_τ CC)**
  Because the tau lepton has an extremely short lifetime ($\tau_\tau \simeq 2.9 \times 10^{-13}$ s), its decay occurs very close to its production point except at very high energies. A ν_τ CC interaction can yield several possible signatures:

  - a hadronic cascade from $\tau \to$ hadrons ($\sim 17\%$),
  - an electromagnetic shower from $\tau \to e$ ($\sim 65\%$),
  - a track-like event from $\tau \to \mu$ ($\sim 17\%$).

  At PeV energies the tau may travel tens of meters before decaying, producing the characteristic "double-bang" signature: one cascade at tau creation and a second at tau decay.

These topology classes form the foundation of neutrino flavor identification and reconstruction in modern Cherenkov telescopes. Their distinct light patterns, combined with the excellent optical clarity of deep-sea water, allow KM3NeT

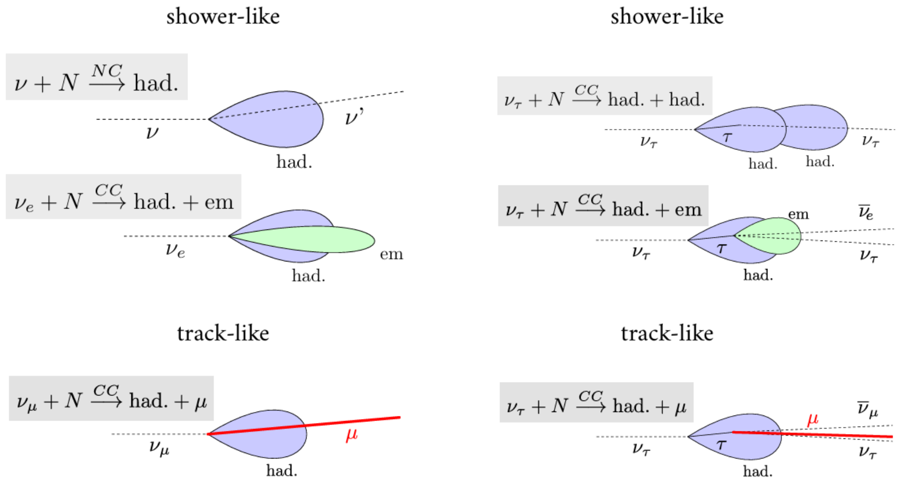

Figure 7: Neutrino Event Topology Drawings by [7]

in particular to achieve sub-degree angular precision and robust separation between track-like and shower-like events across a wide energy range.

4 Pivotal Neutrino Experiments

Experiments illuminate the road for theory and discovery, and this is especially true in the case of neutrino oscillations. There are two main experiments that have played a central role in shaping our current understanding of neutrino oscillations, namely the Super-Kamiokande experiment and the SNO (Sudbury Neutrino Observatory) experiment. In this section, we talk a little bit further about their key findings and how these results provided crucial evidence for neutrino oscillations. We also briefly discuss the reasons behind these experiments being recognized with the Nobel Prize in Physics in 2015, in acknowledgment of their major contributions to our understanding of neutrino properties and behavior.

4.1 Super-Kamiokande (Super-K)

The Super-Kamiokande experiment played a transformative role in neutrino physics by providing the first high-significance evidence for neutrino oscillations. The detector was conceived to address longstanding anomalies from earlier experiments such as the Homestake chlorine detector and the original Kamiokande instrument. Both had observed a persistent deficit in solar electron neutrinos compared to theoretical predictions from the Standard Solar Model, a discrepancy known as the "solar neutrino problem." By the early 1990s, this deficit remained unresolved, with uncertainties as to whether it originated from new neutrino physics or from incomplete solar modelling.

Super-Kamiokande is located in the Kamioka Mine in Japan, approximately 1,000 m underground, where the surrounding rock provides shielding from cosmic-ray backgrounds. The detector consists of a cylindrical stainless-steel tank filled with about 50,000 tons of ultrapure water, viewed by more than 11,000 large photomultiplier tubes (PMTs) that provide nanosecond timing and high-sensitivity photon detection. This configuration enables the precise observation of Cherenkov radiation produced when charged particles travel through water faster than the speed of light in the medium.

Super-K detects neutrinos primarily through elastic scattering processes of the form $\nu + e^- \rightarrow \nu + e^-$, which are especially sensitive to electron neutrinos but retain some sensitivity to muon and tau neutrinos. A key feature of the experiment is its ability to distinguish electron-like events, which produce diffuse Cherenkov rings, from muon-like events, which generate sharper and more well-defined rings.

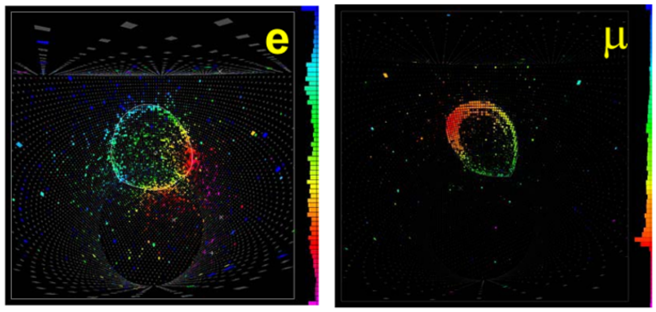

Figure 8: Electron and Muon Cherenkov Signatures [4]

The Super-Kamiokande detector was sensitive to neutrino energies down to about $E_\nu \sim 5$ MeV. Below this energy threshold, background events from β-decay dominate, and therefore Super-Kamiokande was effectively sensitive to the flux of ^{8}B solar neutrinos. The figure below shows the reconstructed direction of electrons with respect to the direction of the Sun for solar neutrinos with energies $E_\nu > 5$ MeV. The theoretical value of the flux that was expected from the Standard Solar Model was around $\Phi_{\text{theory}} \approx 0.45 \pm 0.02$. However, as can be seen from the graph, in the directions pointing directly at the Sun, i.e. at $\cos\theta = 1$, the observed electron neutrino flux was only about $\Phi_e \approx 0.25$, which is roughly half of the theoretically expected value. This discrepancy suggested that there was something nontrivial going on: either a significant fraction of solar electron neutrinos do not arrive at the detector as ν_e, or there is some problem or missing ingredient in the Standard Solar Model itself.

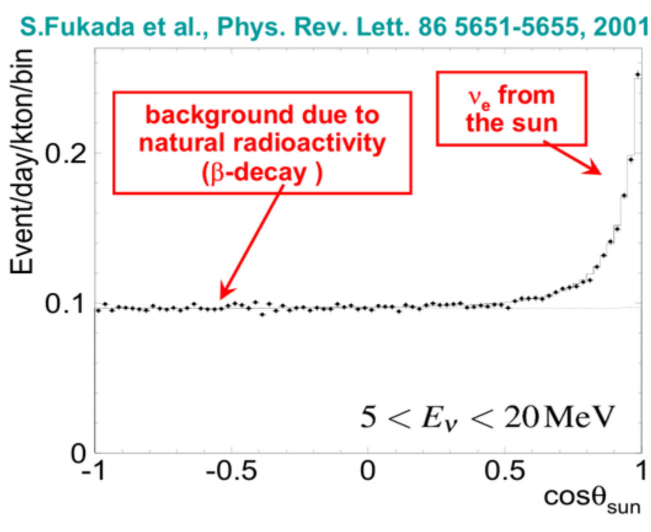

Figure 9: Neutrino data plotted against the reconstructed cosing angle towards the sun [8]

4.2 The Sudbury Neutrino Observatory (SNO)

The Sudbury Neutrino Observatory provided the decisive experimental evidence that resolved the solar neutrino problem by directly demonstrating that electron neutrinos produced in the Sun transform into other flavors during their journey to Earth. Prior to SNO, the observed deficit of solar ν_e could arise either from new neutrino oscillation physics or from incorrect predictions in the Standard Solar Model. No existing detector could simultaneously measure the electron neutrino flux and the total flux of all neutrino flavors. SNO was designed specifically to fill this gap.

Located 2 km underground in the Creighton mine near Sudbury, Canada, SNO used 1,000 tons of heavy water (D_2O) contained within a 12-meter spherical acrylic vessel. Heavy water was essential because it allowed access to three distinct interaction channels on deuterons, each with different flavor sensitivities. Approximately 9,600 PMTs surrounded the vessel, enabling high-precision detection of Cherenkov light.

SNO was the first experiment capable of separately measuring charged-current (CC), neutral-current (NC), and elastic-scattering (ES) interactions. Using these three channels, it could directly compare the surviving electron–neutrino flux to the total solar neutrino flux and thus test the neutrino oscillation hypothesis in a very clean way:

- **Charged-current (CC):** This channel is sensitive exclusively to electron neutrinos and therefore measures the surviving ν_e component coming from the Sun. SNO measured 1968 ± 61 CC events, with a rate proportional to $\phi(\nu_e)$.
- **Elastic scattering (ES):** This channel is predominantly sensitive to electron neutrinos but also to muon and tau neutrinos with a reduced cross section, so it provides a mixed, but still ν_e-dominated, handle on the solar flux. SNO observed 264 ± 26 ES events, with a rate proportional to
$$\phi(\nu_e) + 0.154\,[\phi(\nu_\mu) + \phi(\nu_\tau)].$$
- **Neutral-current (NC):** This channel is equally sensitive to all three active flavors (ν_e, ν_μ, ν_τ), allowing a direct measurement of the total solar neutrino flux, independent of oscillations between flavors. SNO measured 576 ± 50 NC events, with a rate proportional to
$$\phi(\nu_e) + \phi(\nu_\mu) + \phi(\nu_\tau).$$

By comparing these three interaction channels, SNO was able to extract separate fluxes for the different neutrino flavors and directly confront them with the Standard Solar Model (SSM). From their analysis, they found an electron–neutrino flux of
$$\phi(\nu_e) = (1.8 \pm 0.1) \times 10^{-6}\,\mathrm{cm^{-2}\,s^{-1}},$$
while the combined muon- and tau-neutrino flux was
$$\phi(\nu_\mu) + \phi(\nu_\tau) = (3.4 \pm 0.6) \times 10^{-6}\,\mathrm{cm^{-2}\,s^{-1}}.$$
For comparison, the SSM prediction for the electron–neutrino flux at Earth is
$$\phi_{\mathrm{SSM}}(\nu_e) = 5.1 \times 10^{-6}\,\mathrm{cm^{-2}\,s^{-1}}.$$
Thus, SNO demonstrated that while the ν_e flux actually arriving at Earth is significantly lower than the SSM expectation, the total flux of all active neutrino flavors,
$$\phi(\nu_e) + \phi(\nu_\mu) + \phi(\nu_\tau),$$
agrees very well with the solar model. This comparison provided direct and essentially model-independent evidence that solar electron neutrinos undergo flavor transformation (i.e. neutrino oscillations) as they propagate from the Sun to the detector.

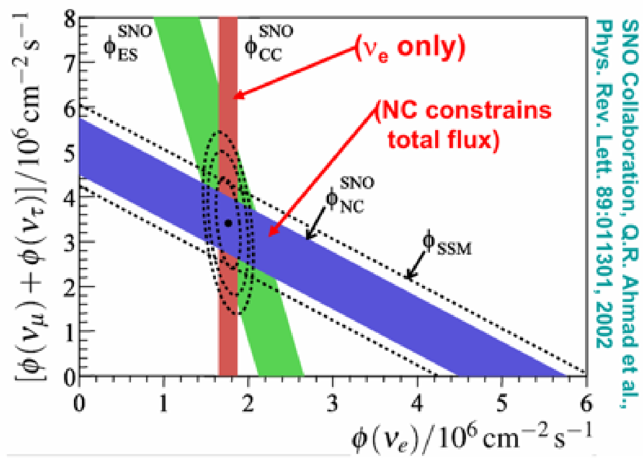

Figure 10: Neutrino Flux of different flavors from different interaction processes [9]

5 Mass Order Hierarchy

5.1 Do Neutrinos Have Mass? Evidence and Tension with the Standard Model

In the original formulation of the Standard Model (SM), neutrinos were assumed to be strictly massless. In this picture the SM predicted three massless neutrino species that retain their flavor identity during propagation. However, a series of experiments, some of which have been discussed in this report (such as Super-Kamiokande and SNO, among many others), have shown very clearly that neutrinos change flavor as they propagate. This phenomenon of flavor conversion, or neutrino oscillation, can only occur if neutrino flavor states are quantum-mechanical mixtures of different mass eigenstates that evolve with different phases in time. Thus, the experimental observation of neutrino oscillations implies in a direct way that at least two neutrino mass eigenstates have non-zero mass, and therefore neutrinos cannot be massless as originally assumed in the SM.

Although we now know neutrinos have mass, measuring those masses directly is extremely challenging. Neutrinos interact only via the weak interaction and their masses are tiny, so kinematic effects in processes such as β decay are very small. As a result, current laboratory measurements can only place upper limits on the absolute values of the neutrino masses. Typical bounds [4] are

$$m_{\nu_e} < 1 \text{ eV}, \qquad m_{\nu_\mu} < 0.17 \text{ MeV}, \qquad m_{\nu_\tau} < 18.2 \text{ MeV},$$

which tell us that neutrinos are extremely light compared to all the charged leptons, but do not yet pin down their exact masses.

5.2 Mass Limits and Why Oscillation Experiments Measure Only Δm^2

Neutrino oscillation experiments are sensitive not to the individual masses themselves, but rather to differences in the squared masses of the neutrino mass eigenstates. This arises because the flavor–transition probabilities we observe and estimate rely on those square mass differences:

$$\Delta m_{21}^2 = m_2^2 - m_1^2, \qquad \Delta m_{31}^2 = m_3^2 - m_1^2$$

Two independent mass-squared splittings have been determined with good precision. Solar neutrino experiments probe the smaller "solar" mass splitting,

$$\Delta m_{21}^2 \approx 7.4 \times 10^{-5}\ \mathrm{eV}^2,$$

which is associated with long-wavelength oscillations over astronomical distances. Atmospheric and accelerator neutrino experiments are sensitive to the larger "atmospheric" mass splitting,

$$|\Delta m_{32}^2| \approx 2.5 \times 10^{-3}\ \mathrm{eV}^2,$$

corresponding to shorter-wavelength oscillations over terrestrial baselines. The large separation between these two scales reflects the characteristic two-frequency structure of neutrino oscillations. [5]

Because oscillation experiments depend only on these Δm^2 splittings, they cannot by themselves determine the absolute neutrino mass scale. Instead, the absolute masses must be constrained by non-oscillation techniques leaving the full neutrino mass spectrum still partly unknown.

5.3 Normal vs. Inverted Mass Hierarchy

Although the mass splittings are known, the ordering of the neutrino mass eigenstates remains uncertain. Two possibilities exist. In the *normal hierarchy* (NH),

$$m_1 < m_2 < m_3,$$

the pattern resembles that of charged leptons and quarks. In the *inverted hierarchy* (IH),

$$m_3 < m_1 < m_2,$$

the heaviest and lightest states are exchanged. The sign of the atmospheric mass splitting determines the hierarchy, but current data do not yet conclusively favor one ordering.

Determining the hierarchy has major implications.In cosmology, the hierarchy influences how neutrino masses suppress small-scale structure formation. Supernova neutrino flavor evolution depends sensitively on the ordering, as do Earth matter effects observed in atmospheric neutrinos. The hierarchy also affects measurements of the leptonic CP-violating phase, which may play a role in leptogenesis and the matter–antimatter asymmetry of the Universe.

A new generation of experiments such as **KM3NeT/ORCA** aims to resolve the hierarchy with high significance by exploit matter effects in the Earth.

5.4 Flavor States vs. Mass States: Linear Superposition

The discovery of oscillations revealed that neutrinos produced in weak interactions are *flavor eigenstates*, while the states that propagate are *mass eigenstates*. A neutrino state can be written as a linear combination in either the flavor or mass basis. These two bases are connected by a unitary mixing matrix U [10] such that

$$\begin{pmatrix} |\nu_e\rangle \\ |\nu_\mu\rangle \\ |\nu_\tau\rangle \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} \begin{pmatrix} |\nu_1\rangle \\ |\nu_2\rangle \\ |\nu_3\rangle \end{pmatrix}.$$

Because the mass eigenstates have different masses, they accumulate different quantum phases as they propagate. The relative phase differences lead to interference effects, modifying the probabilities of detecting each flavor at a given distance.

If all neutrino masses were identical, these phases would evolve in exactly the same way and no oscillation would occur. Therefore, the phenomenon of oscillation is possible only if neutrinos have distinct nonzero masses. The interference between the evolving mass eigenstates provides the physical origin of flavor oscillation.

5.5 The PMNS Matrix: Structure, Parameters, and Unitarity

The transformation matrix U is called the Pontecorvo–Maki–Nakagawa–Sakata (PMNS) matrix and describes the mixing between flavor and mass eigenstates. Assuming it is a unitary matrix, $U^\dagger U = I$, it ensures probability conservation and the orthogonality of the flavor states. A convenient parametrization of U in terms of three mixing angles and one CP-violating phase is

$$\begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix}}_{\text{“atmospheric”}} \underbrace{\begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix}}_{\text{reactor/CP}} \underbrace{\begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{“solar”}},$$

where $c_{ij} \equiv \cos\theta_{ij}$ and $s_{ij} \equiv \sin\theta_{ij}$.

The PMNS matrix contains three mixing angles—θ_{12}, θ_{23}, and θ_{13}—and one Dirac CP-violating phase, δ_{CP}. Each angle governs specific oscillation channels: θ_{12} predominantly controls solar oscillations, θ_{23} dominates atmospheric oscillations, and θ_{13} is most relevant for reactor-based oscillation experiments. A variety of experiments isolate these parameters: solar experiments such as SNO and Super-Kamiokande determine Δm^2_{21} and θ_{12}, reactor experiments such as Daya Bay constrain θ_{13}, and atmospheric and long-baseline accelerator experiments probe $\left|\Delta m^2_{32}\right|$ and θ_{23}.

The Dirac phase δ_{CP} remains one of the least well-known parameters and could reveal CP violation in the lepton sector, a key ingredient in many theories of leptogenesis and the matter–antimatter asymmetry of the Universe.

5.6 Oscillation Probabilities and Wavelength Scales

The oscillation probability for a neutrino of flavor α to convert into flavor β after traveling a distance L with energy E can be written in a very compact way [10] in terms of the mixing matrix elements and the mass splittings. In general we have

$$P_{\alpha\to\beta}(L, E) = \sum_{i,j} U_{\alpha i}\, U^*_{\beta i}\, U^*_{\alpha j}\, U_{\beta j} \exp\left(-i\, \frac{\Delta m^2_{ij} L}{2E}\right),$$

where

$$\Delta m^2_{ij} \equiv m_i^2 - m_j^2$$

is the squared mass difference between the mass eigenstates ν_i and ν_j.

A useful quantity that appears naturally in this expression is the characteristic oscillation length,

$$L_{\text{osc}} \sim \frac{4\pi E}{\Delta m^2},$$

which sets the wavelength of the flavor transitions. Because the solar mass splitting Δm^2_{21} is relatively small, solar neutrinos probe long-wavelength oscillations. In contrast, the larger atmospheric splitting Δm^2_{32} leads to short-wavelength oscillations that are relevant for atmospheric and accelerator neutrinos.

By imposing unitarity on the PMNS matrix, taking it to be real (neglecting CP violation for simplicity), and using the approximation $\Delta m^2_{31} \simeq \Delta m^2_{32}$, one can derive a convenient approximate expression for the electron–neutrino survival probability. In this limit one finds

$$P(\nu_e \to \nu_e) \simeq 1 - \cos^4\theta_{13}\,\sin^2 2\theta_{12}\,\sin^2\Delta_{21} - \sin^2 2\theta_{13}\,\sin^2\Delta_{32}, \tag{1}$$

where the dimensionless phases

$$\Delta_{ij} \equiv \frac{\Delta m^2_{ij} L}{4E}$$

encode the dependence on the baseline and the energy.
For a neutrino with energy $E = 1$ MeV, the corresponding oscillation wavelengths are roughly

$$\lambda_{21} \simeq 30 \text{ km}, \qquad \lambda_{32} \simeq 0.8 \text{ km},$$

illustrating the presence of both a long-wavelength (solar) and a short-wavelength (atmospheric) component in the electron–neutrino survival probability. The figure below shows how these oscillation modes combine to give the survival probability of ν_e as a function of the propagation distance.

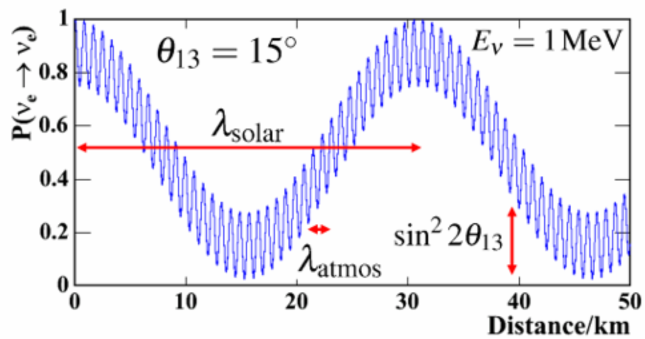

Figure 11: Electron Neutrino Survival Probability across distance in km [4]

6 The KM3NeT Experiment

6.1 Overview of the KM3NeT Collaboration

KM3NeT is a major European deep-sea neutrino experiment designed to advance our understanding of both **neutrino oscillation physics** and **high-energy astrophysical neutrinos**. It consists of two large underwater detectors located in the Mediterranean Sea: **ORCA**, near Toulon (France), and **ARCA**, near Sicily (Italy). Together, these installations instrument several cubic kilometres of deep seawater with thousands of optical sensors [11].
The project began construction in the mid-2010s, following the success of the ANTARES experiment. Since then, KM3NeT has made rapid progress. Key milestones include the deployment of initial detection units, completion of seafloor infrastructures, and the achievement of stable long-term data-taking. By 2024–2025, ORCA exceeded thirty deployed Detection Units (DUs), while ARCA passed fifty DUs, marking significant advancement toward the full infrastructure. The collaboration anticipates completing all three planned building blocks by the late 2020s or around 2030.
The collaboration consists of hundreds of scientists from more than a dozen countries. The Mediterranean location offers several advantages: long optical scattering and absorption lengths, excellent angular resolution for track events, and the logistical benefit of direct sea access for deployment and maintenance—unlike ice-based observatories.
To accomplish its goals, KM3NeT employs a modular architecture centred around three important components:

- **Building Block:** A large-scale structural unit containing 115 Detection Units.
- **Detection Unit (DU):** A vertical string with 18 Digital Optical Modules, anchored to the seabed and supported by a buoy.
- **Digital Optical Module (DOM):** A pressure-resistant sphere containing 31 photomultiplier tubes with nanosecond-level timing.

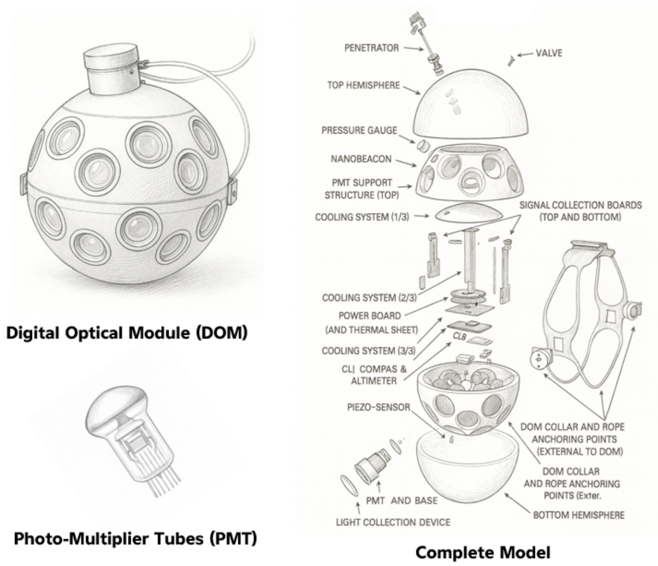

Figure 12: Sketches of the Digital Optical Module & Photo-Multiplier Tubes adapted from [11]

6.2 ORCA: Oscillation Research with Cosmics in the Abyss

ORCA is optimised for **low-energy neutrinos** ranging from a few GeV to approximately 30 GeV. Its principal objective is to determine the **neutrino mass hierarchy** (normal or inverted) by analysing matter effects on atmospheric neutrinos as they traverse the Earth. ORCA also aims to precisely measure oscillation parameters such as θ_{23}, Δm^2_{32}, and the impact of the Earth's density profile.

Atmospheric neutrinos act as a natural beam, providing a continuous spectrum of energies and baselines. Up-going neutrinos, which pass through the Earth before detection, experience strong matter-induced modifications to their oscillation probabilities. These effects differ between the normal and inverted mass hierarchies, forming the core of ORCA's sensitivity.

The ORCA detector is installed at depths of roughly 2.5–3.5 km. Its geometry features:

- **Dense vertical spacing** of roughly 9 m between DOMs,
- **Closely packed three-dimensional array** ideal for reconstructing faint Cherenkov signatures,
- **Sufficient granularity** for accurate energy and direction reconstruction at GeV energies.

Some of the main challenges faced by ORCA include the accurate reconstruction of low-energy events, efficient rejection of downward-going atmospheric muons, and the mitigation of systematic uncertainties related to seawater optical properties and environmental backgrounds. Early calibration campaigns and initial analyses confirm ORCA's excellent performance.

Reconstruction and classification techniques rely heavily on both traditional methods (likelihood-based fits) and modern machine learning tools, including boosted decision trees, convolutional neural networks (CNNs), and graph neural networks (GNNs).

6.3 ARCA: Astroparticle Research with Cosmics in the Abyss

ARCA is designed as the high-energy counterpart of KM3NeT, targeting neutrinos in the **TeV to PeV energy range**. Its primary mission is to identify and study astrophysical neutrino sources, including active galactic nuclei (AGN), gamma-ray bursts (GRBs), supernova remnants (SNRs), and other extreme cosmic accelerators capable of producing ultra-high-energy cosmic rays.

High-energy neutrinos produce long muon tracks and bright cascades, making large detection volumes essential. ARCA achieves this by using a sparse DOM configuration spanning approximately **one cubic kilometre** of deep seawater. This geometry is optimized for reconstructing extended muon tracks with **exceptional angular resolution**, often better than 0.2° at high energies.

ARCA is integral to **multi-messenger astronomy**. It provides rapid alerts for candidate astrophysical neutrino events, enabling coordinated observations with gamma-ray observatories such as Fermi-LAT, MAGIC, and CTA, as well as gravitational-wave detectors like LIGO, Virgo, and KAGRA. The clarity of seawater gives ARCA a sharper pointing capability than ice-based detectors such as IceCube, particularly for muon neutrino events.

Scientific analyses include point-source searches, diffuse astrophysical flux measurements, and time-dependent studies of transient events. Distinguishing cosmic neutrinos from atmospheric backgrounds relies on directional and energy reconstruction, supplemented by machine-learning classifiers. Future developments are expected to include real-time deep-learning-based event tagging for instantaneous astrophysical alerts. Early in 2025, an exciting anomaly of high energy muon around 120 PeV was detected by ARCA which suggest the detection of a neutrino far beyond the solar and atmospheric neutrino energies we detect [12].

6.4 Neutrino Detection Principles in KM3NeT

Neutrino detection in KM3NeT is based on the measurement of **Cherenkov radiation** emitted when charged particles produced in neutrino interactions travel faster than the local speed of light in seawater. This radiation forms a **coherent cone of blue light**, allowing the direction and energy of the originating particle to be reconstructed. The deep Mediterranean's excellent optical clarity and long scattering length significantly enhance reconstruction performance. Neutrinos interacting in or near the detector volume can produce several characteristic event topologies:

- **Track-like events**, typically from ν_μ charged-current (CC) interactions, which generate long muon tracks that traverse the detector and provide precise directional information.
- **Shower-like (cascade) events** from ν_e CC, ν_τ CC at sub-PeV energies, or neutral-current (NC) interactions of any flavour. These produce compact light patterns with good energy resolution.
- **Double-bang events**, characteristic of high-energy ν_τ CC interactions, in which a tau lepton is produced and later decays, generating two spatially separated showers.

Each Digital Optical Module contains 31 small photomultiplier tubes that measure photon arrival times with nanosecond accuracy and record the number of detected photons. Reconstruction algorithms combine timing triangulation, maximum-likelihood fits, and machine-learning models such as CNNs and GNNs to infer the neutrino flavour, trajectory, and energy.

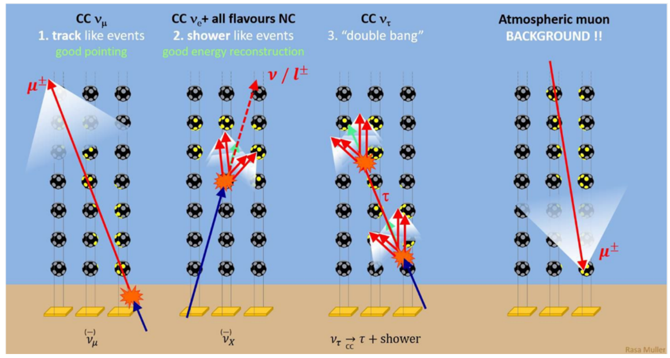

Figure 13: Neutrino Event Topolgy detection by KM3NeT DOMs [13]

Systematic effects—including variations in water optical properties, DOM calibration drifts, bioluminescent activity, and natural radioactivity—are continuously monitored and corrected. Background suppression is essential, as downward-going atmospheric muons vastly outnumber neutrino-induced events. KM3NeT employs a combination of upward-going event selection, topological discriminants, and machine- learning classifiers to isolate a clean neutrino sample.

7 Broader Physics Research Opportunities

At the heart of every physics research endeavor, we find connections between phenomena we did not anticipate. The further study and detection of neutrinos through KM3NeT and other experiments may shed light—often indirectly—on open problems in physics [11].

- **Gamma-ray bursts**

  Gamma-ray bursts (GRBs) are well known as potential cosmic re-acceleration sites via diffusive shock acceleration inside their jets. Because GRBs are short-lived (seconds to minutes), time-tagged searches around a burst can also target coincident neutrinos within a narrow window, reducing background from steady point sources. If neutrinos are observed in temporal and directional coincidence with GRBs, they would probe hadronic processes in the jets and the nature of particle acceleration at extreme energies.

- **Cosmic-ray physics**

  A major source of atmospheric neutrinos comes from cosmic-ray interactions in the atmosphere, producing pions that decay into muons and neutrinos. Neutrino experiments' ability to detect atmospheric neutrinos (and, as a by-product, atmospheric muons) gives us a window into the spectrum, composition, and interaction models of the primary cosmic rays. These measurements can also inform studies of the large-scale anisotropy seen in cosmic-ray maps and help test hadronic interaction models at energies beyond accelerator reach.

- **Particle physics beyond the Standard Model**

  As mentioned earlier, improved measurements of the PMNS mixing matrix and mass-splitting parameters can sharpen our picture of neutrino properties and test whether additional particles or interactions are needed beyond the Standard Model. As new theories develop, we can confront them with data—either ruling them out or gaining support—and iteratively move toward a more accurate description.

  Another interesting area is the production of charm quarks in high-energy cosmic-ray interactions in the atmosphere. Their prompt decays to leptons can generate a "prompt" atmospheric neutrino component at high energies. Identifying and measuring muons and neutrinos from these processes would clarify the underlying reaction mechanisms and help separate prompt from conventional atmospheric backgrounds.

The continuous measurement of high-energy neutrinos may also reveal events at energies beyond simple expectations. When such outliers appear, they often push us to revise models of sources, acceleration, and propagation to understand how those neutrino energies are achieved.

- **Dark matter**

  While dark matter is well established through gravitational evidence, we know very little about its particle nature because it does not couple to electromagnetic or strong forces. Some models predict that if dark-matter particles have masses in the GeV–TeV range, their annihilation or decay could produce neutrinos. Neutrino telescopes therefore search for excess neutrino fluxes from regions with high dark-matter density—such as the Galactic Center, the halo, or even the Sun—to place constraints on dark-matter properties. Different experiments have already set competitive limits by studying these fluxes, and continued observations will keep tightening those constraints.

References

[1] Wolfgang Pauli. "Dear radioactive ladies and gentlemen". In: *Physics Today* 31.9 (1978), p. 27.

[2] Enrico Fermi. "Versuch einer Theorie der β-Strahlen. I". In: *Zeitschrift für Physik* 88 (1934), pp. 161–177. ISSN: 0044-3328. DOI: 10.1007/BF01351864. URL: https://doi.org/10.1007/BF01351864.

[3] F. Reines et al. "Detection of the Free Antineutrino". In: *Physical Review* 117 (1960), pp. 159–173. DOI: 10.1103/PhysRev.117.159. URL: https://link.aps.org/doi/10.1103/PhysRev.117.159.

[4] Mark Thomson. *Handout 11: Neutrino Oscillations*. Lecture Handout, Part III Particle Physics, University of Cambridge. Accessed 23 Nov 2025. 2009. URL: https://www.hep.phy.cam.ac.uk/~thomson/lectures/partIIIparticles/Handout11_2009.pdf.

[5] Mark Thomson. *Modern Particle Physics*. Cambridge, UK: Cambridge University Press, 2013. ISBN: 9781107034266.

[6] J. N. Bahcall and M. H. Pinsonneault. "What Do We (Not) Know Theoretically about Solar Neutrino Fluxes?" In: *Physical Review Letters* 92.12 (2004), p. 121301. DOI: 10.1103/PhysRevLett.92.121301. URL: https://doi.org/10.1103/PhysRevLett.92.121301.

[7] Steffen Hallmann. "Sensitivity to atmospheric tau-neutrino appearance and all-flavour search for neutrinos from the Fermi Bubbles with the deep-sea telescopes KM3NeT/ORCA and ANTARES". en. Available under CC BY-NC-SA 4.0. Doctoral thesis. Erlangen, Germany: Friedrich-Alexander-Universität Erlangen-Nürnberg, Feb. 2021. URL: https://open.fau.de/handle/openfau/15749.

[8] S. Fukuda and Super-Kamiokande Collaboration. "Solar $^{8}\mathrm{B}$ and hep Neutrino Measurements from 1258 Days of Super-Kamiokande Data". In: *Physical Review Letters* 86.25 (2001), pp. 5651–5655. DOI: 10.1103/PhysRevLett.86.5651. URL: https://doi.org/10.1103/PhysRevLett.86.5651.

[9] Q. R. Ahmad and SNO Collaboration. "Direct Evidence for Neutrino Flavor Transformation from Neutral-Current Interactions in the Sudbury Neutrino Observatory". In: *Physical Review Letters* 89 (2002), p. 011301. DOI: 10.1103/PhysRevLett.89.011301. URL: https://doi.org/10.1103/PhysRevLett.89.011301.

[10] Lukas Hennig. “Tau neutrino identification with Graph Neural Networks in KM3NeT/ORCA”. Master’s thesis. MA thesis. Friedrich-Alexander-Universität Erlangen–Nürnberg (FAU), June 2023. URL: https://ecap.nat.fau.de/wp-content/uploads/2023/06/2023-06_LukasHennig_MSc_ORCA_TauID_GNNs.pdf.

[11] S. Adrián-Martínez et al. “Letter of intent for KM3NeT 2.0”. In: *Journal of Physics G: Nuclear and Particle Physics* 43.8 (2016), p. 084001. DOI: 10.1088/0954-3899/43/8/084001. arXiv: 1601.07459.

[12] KM3NeT Collaboration. “Observation of an ultra-high-energy cosmic neutrino with KM3NeT”. In: *Nature* 638.8050 (2025), pp. 376–382. DOI: 10.1038/s41586-024-08543-1. URL: https://www.nature.com/articles/s41586-024-08543-1.

[13] Jorge Prado González. *An Overview of Machine Learning Techniques in KM3NeT*. Presentation, Neutrino Physics and Machine Learning, ETH Zurich. Accessed 23 Nov 2025. 2024. URL: https://indico.phys.ethz.ch/event/113/contributions/843/attachments/517/1112/jprado_NPML_presentation_june2024.pdf.